BAHIR DAR UNIVERSITY

Faculty Of Computing Department Of

Software engineering

Operating System and System Programming(OSSP)

Individual assignment

OSSP_individual_Hiwot Adane Mesganaw_BDU1604413_B

Part 1

A. Introduction(Background and Motivation)

An operating system acts as a bridge between computer hardware and the user, managing resources and enabling interaction with the system. However, installing a new operating system directly on physical hardware can be risky, especially for learning or experimentation. Users may accidentally lose important data, corrupt the boot system, or encounter hardware compatibility issues.

Virtualization addresses these challenges by allowing multiple operating systems to run safely on a single physical machine. Tools such as Oracle VM VirtualBox enable users to create virtual machines that simulate real computers within an existing operating system.

The motivation of this assignment is to understand how to install and manage an operating system in a virtual environment. This helps students gain practical knowledge used in real-world IT environments such as system administration, software testing, and cloud computing.

Zorin OS is a modern Linux distribution designed to be user-friendly and familiar to Windows and macOS users. It is based on Ubuntu Long Term Support (LTS), ensuring stability, security updates, and long-term reliability. This makes it an ideal choice for learning and experimentation in a virtual environment.

B. Objectives

- To understand the concept of virtualization and its role in modern computing
- To install Zorin OS on a virtual machine using Oracle VM VirtualBox
- To configure virtual machine resources such as RAM, CPU, and storage for optimal performance
- To create and configure a user account using the name *Hiwot Adane*
- To document the installation process with appropriate screenshots
- To identify and resolve issues encountered during installation
- To analyze system requirements and files ystem support
- To evaluate the advantages and disadvantages of virtualization
- To explain virtualization, including what it is, why it is used, and how it works

C. Requirements

i. Hardware Requirements:

- Minimum 4 GB total (2 GB for VM) RAM and maximum 8 GB total (allocated 4 GB to VM)

- Minimum 64-bit processor CPU and maximum Intel Core i5 (supports VT-x)
- Minimum 20 GB free space storage and maximum 256 GB SSD (allocated 30.29 GB to VM)

ii. Software Requirements:

- Host operating system(Window 11) for the purpose of run vitualBox on my physical machine
- Oracle VM VirtualBox
- Zorin OS IS
-

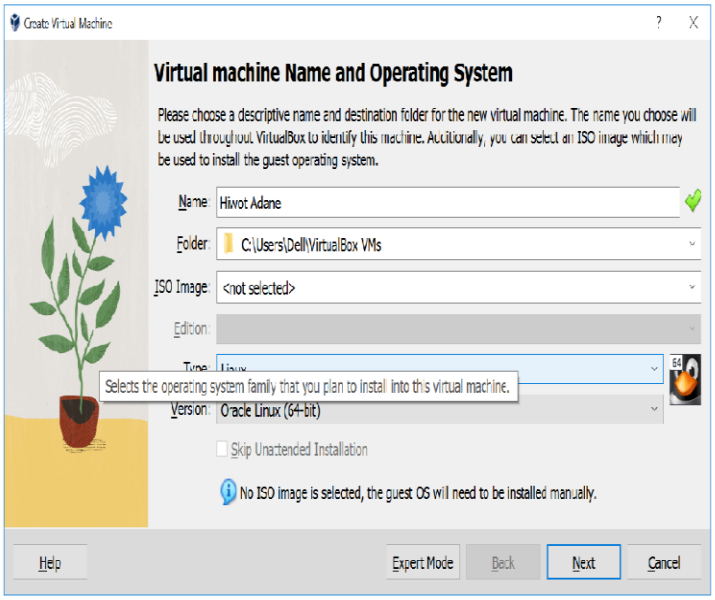
D. Installation Steps

Step 1: Install VirtualBox

- Download and install VirtualBox
- Launch the application

Step 2: Create Virtual Machine

- Click **New**
- Name: *Hiwot-Adane-VM*
- Type: Linux
- Version: oracle linux(64-bit)ff



Step 3: Configure Hardware Resources

In the "Hardware" configuration window, I set:

- Base Memory (RAM): 4096 MB (4 GB)
- Processors: 2 CPUs (changed from 1 to 2 for better performance)
- Enable EFI: Left unchecked (Zorin works fine with BIOS mode)

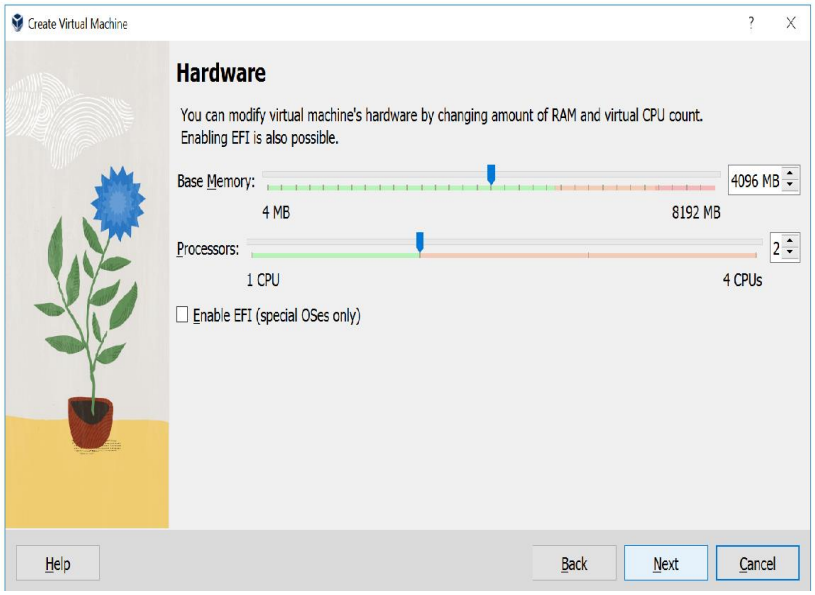


Figure 3*:* Configuring RAM (4 GB) and CPU (2 cores) for the VM

Step 4: Create Virtual Hard Disk

I chose to create a new virtual hard disk with:

- Disk Size:30.29 GB
- Pre-allocate Full Size: Not checked (dynamically allocated saves space)

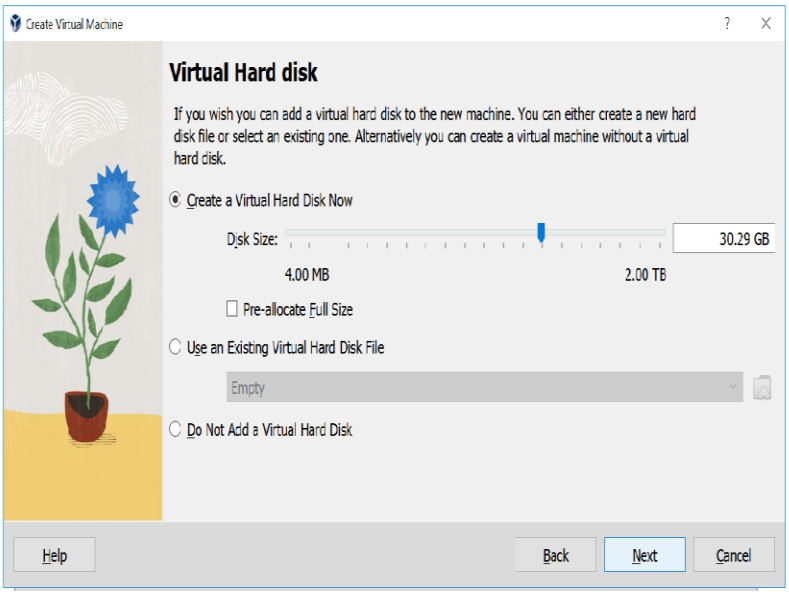


Figure 4*:* Creating a dynamically allocated virtual hard disk of 30.29 GB

Step 5: Review and Create the Virtual Machine

The "Summary" screen showed my complete configuration:

setting	value
Machine Name	Hiwot Adane
Machine Folder	C:/Users/Dell/VirtualBox VMs/Hiwot Adane
Guest OS Type	Oracle Linux (64-bit)
Base Memory	4096 MB
Processor(s)	2
Disk Size	30.29 GB

I clicked **Finish** to create the VM.

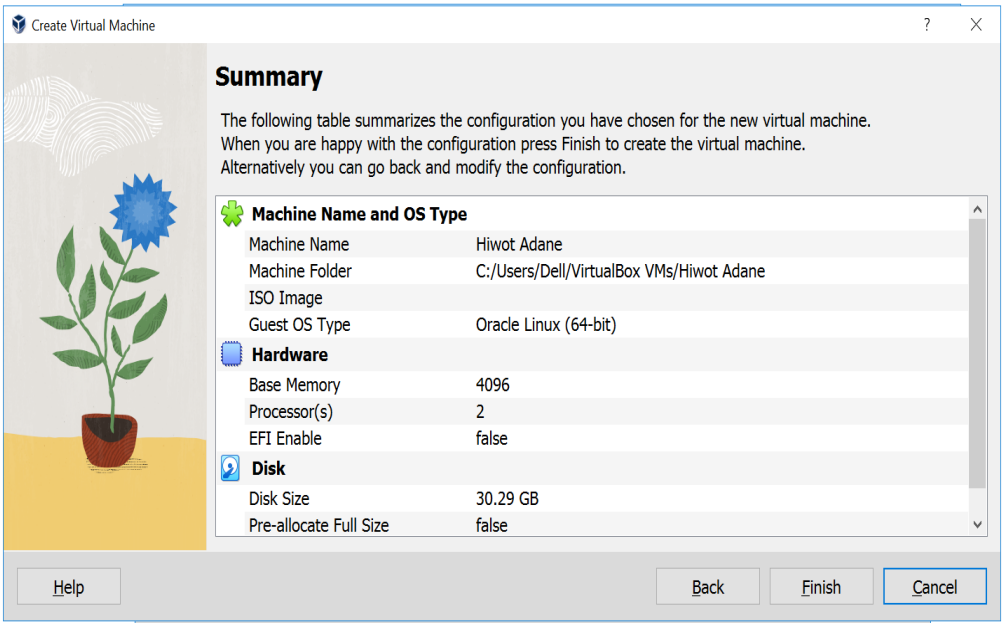


Figure 5: *Summary of VM configuration before creation*

Step 6: VM Created — Initial State

The VirtualBox Manager now showed my VM named **"Hiwot Adane"** with status "Powered Off."

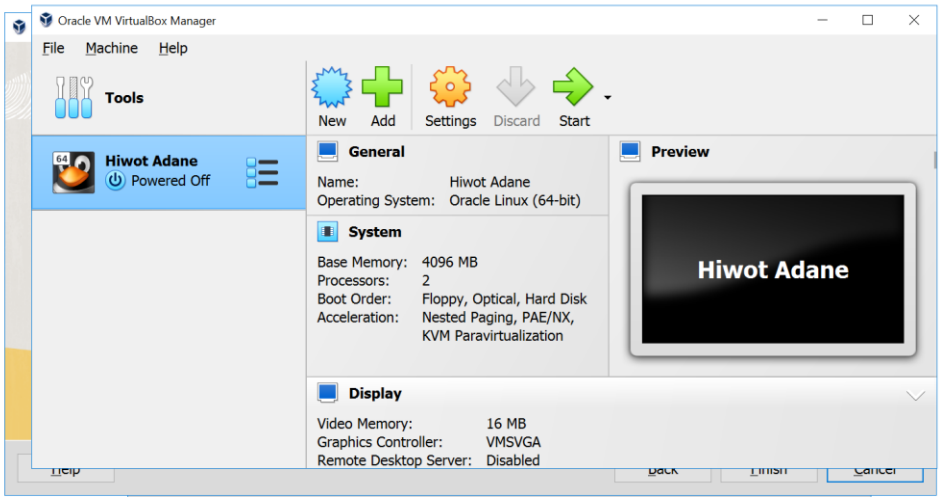


Figure 6: Virtual machine created and ready to start

Step 7: Configure Display Settings (Before Installation)

I noticed the default video memory was only 16 MB, which might cause display issues. I went to **Settings → Display** and increased:

- Video Memory:128 MB
- Graphics Controller: VMSVGA
- Enabled 3D Acceleration: Checked

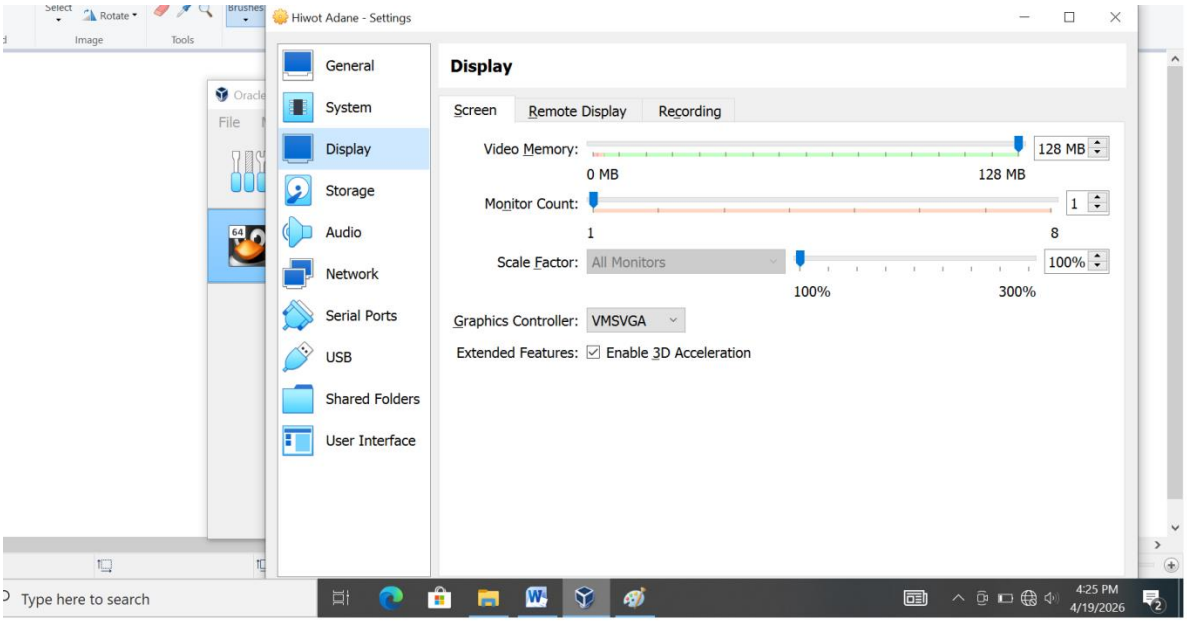
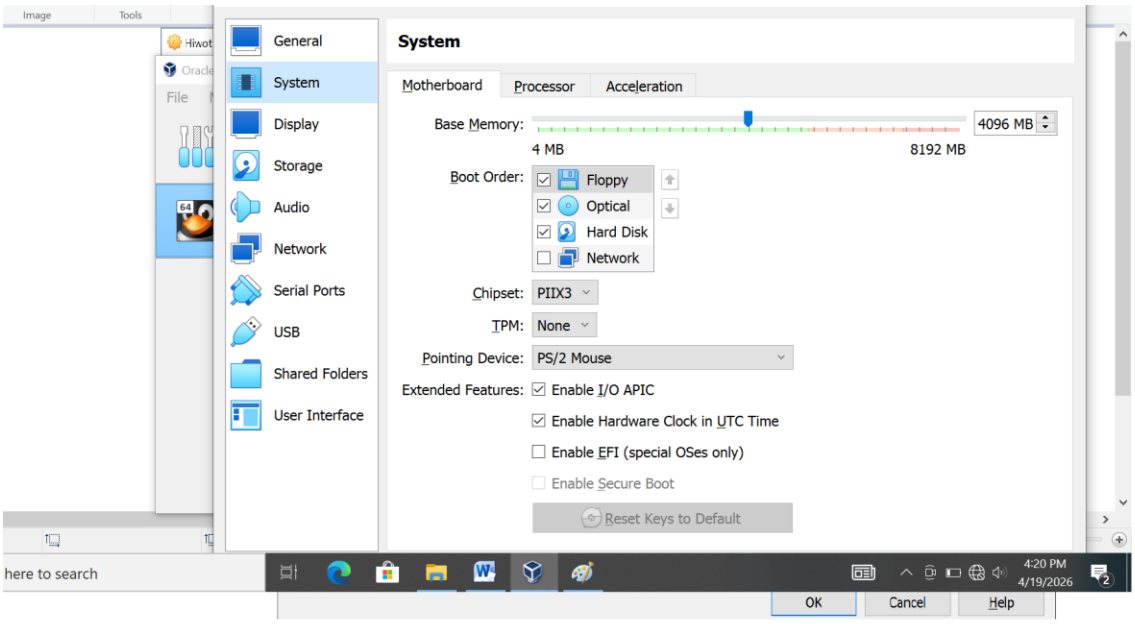


Figure 7: Increasing video memory to 128 MB for better display performance

Step 8: Configure System (Motherboard) Settings

In **Settings → System**, I reviewed the motherboard configuration:

- Base Memory: 4096 MB
- Boot Order: Optical, Hard Disk
- Chipset: PIIX3
- Pointing Device: PS/2 Mouse



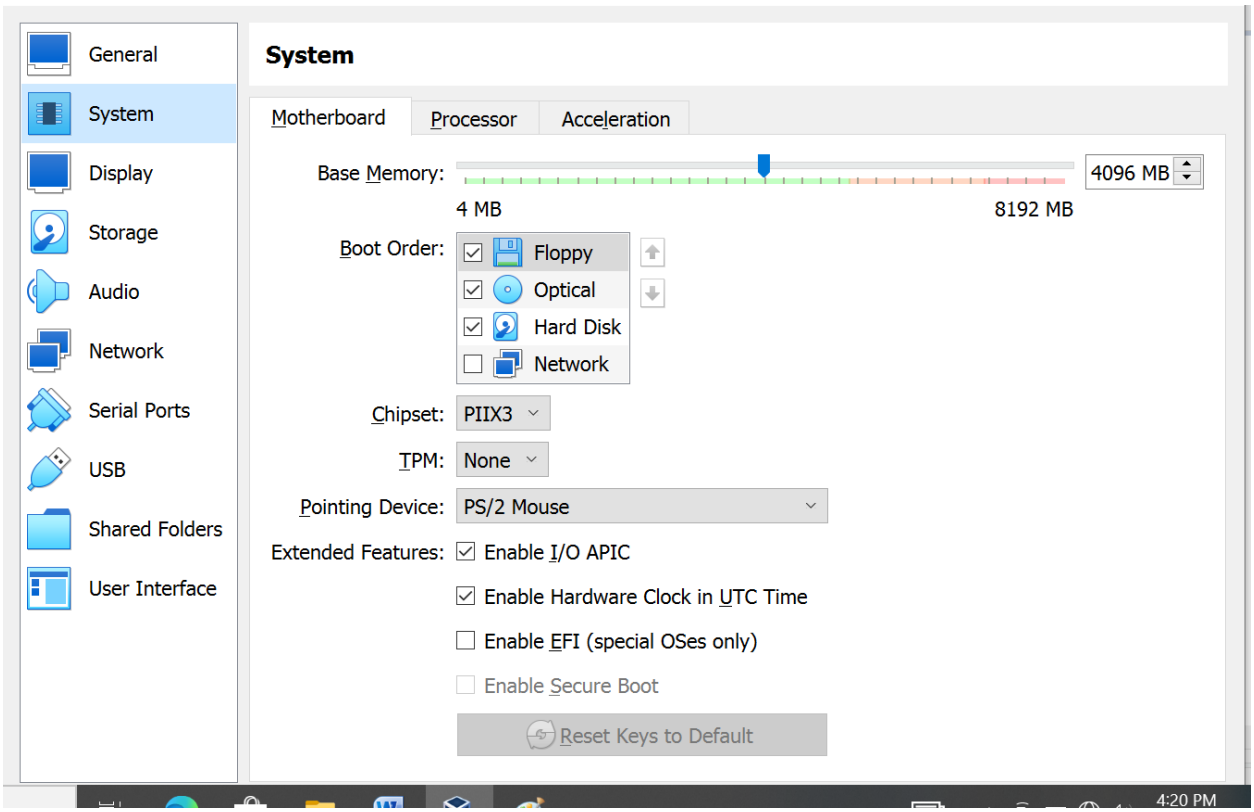


Figure 8 & 9: System and motherboard configuration settings

Step 9: Configure Storage — Attach Zorin OS ISO

In **Settings** → **Storage**, I saw:

- Controller: IDE – Optical Drive (Empty)
- Controller: SATA –Hiwot Adane.vdi (my virtual hard disk)

To attach the Zorin OS ISO, I clicked the empty optical drive → **Choose a disk file** → selected my downloaded Zorin-os-18.1-64-bit.iso

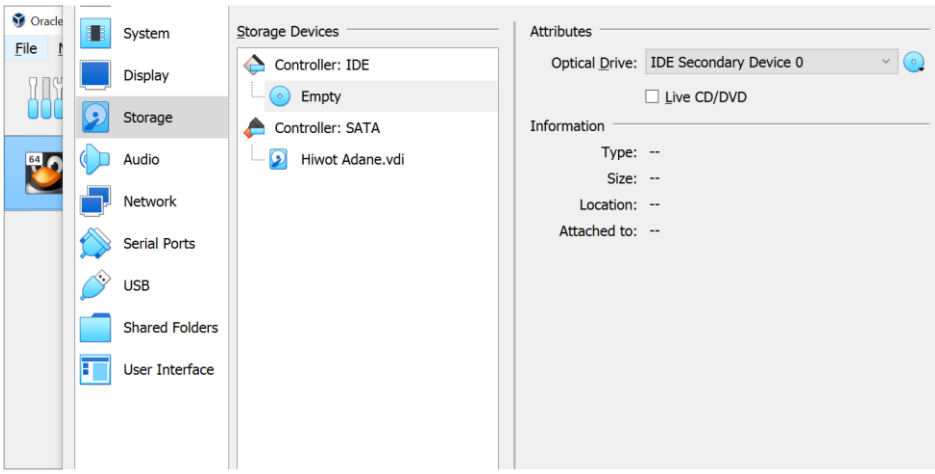


Figure 10: Storage configuration showing the virtual hard disk and empty optical drive ready for ISO

Step 10: Installation process

Screen Name	selection
Boot Menu	Selected "Install Zorin OS" (not "Try Zorin OS")
Language	English (United States)
Keyboard Layout	English (US)
Updates & Software	"Normal installation" + "Download updates while installing Zorin OS"
Installation Type	**"Erase disk and install Zorin OS"** — appropriate for a virtual machine with no existing data
Partitioning	Automatic partitioning by installer. Result: single **ext4** partition mounted at / (root) + swap partition
Time Zone	Selected my location on the interactive world map

User Account	• **Your name:** Hiwot Adane • **Username:** hiwotadane • **Password:** [set a strong password] • **Login:** Require password to log in
Installation Progress	Progress bar advanced over approximately 8-10 minutes while files were copied to the virtual hard disk
Installation Complete	"Installation Complete" dialog appeared → Clicked "Restart Now" → Removed installation ISO when prompted

Partitioning: The Zorin OS installer automatically partitioned the virtual hard disk with:

- One **ext4** partition mounted at / (root) – approximately 29 GB
- One **swap** partition – approximately 1 GB

Why ext4? Zorin OS (like most Linux distributions) uses **ext4** as its default file system because:

- Journaling protects against data corruption after crashes
- Excellent performance on both HDD and SSD
- Supports large files (up to 16 TB) and large volumes (up to 1 EB)
- Backward compatible with ext2 and ext3
- Open source with no licensing restrictions

Once the installation completed and the VM restarted, the Zorin OS login screen appeared

Step 11: Start the Virtual Machine

I selected the **Hiwot Adane** VM and clicked **Start**. The VM booted from the Zorin OS ISO, and after installation completed, the Zorin OS interface appeared.

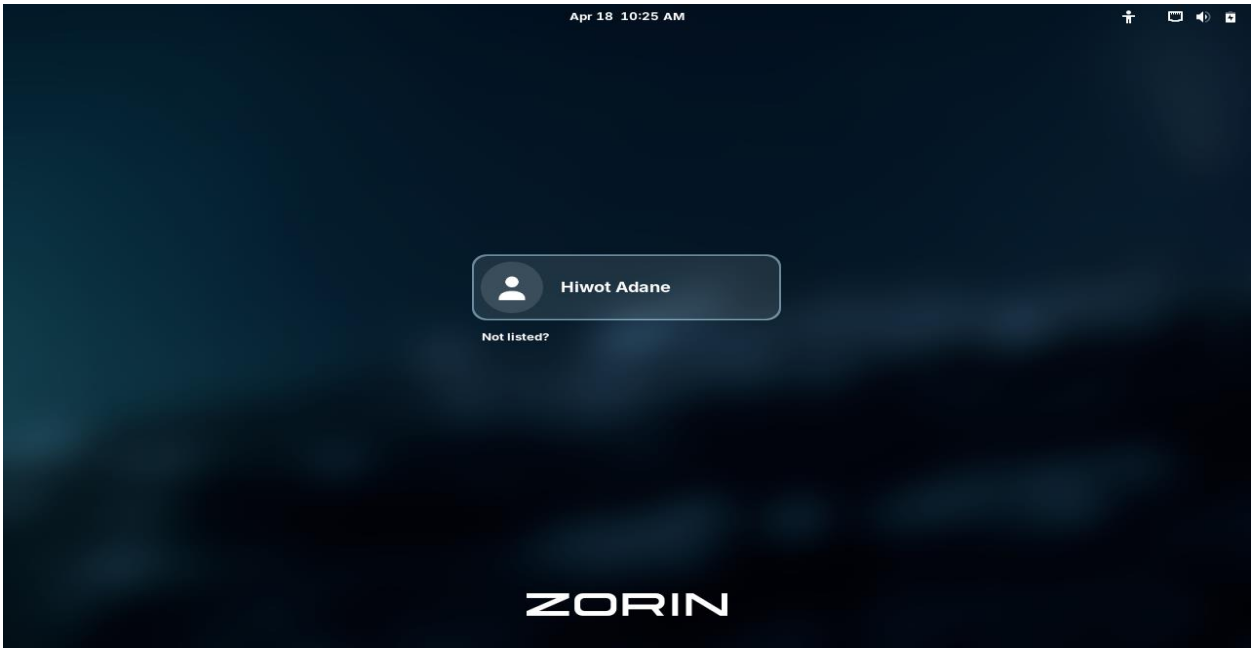


Figure 11: Zorin OS running in the virtual machine after successful installation

Step 12: User Account Creation

This step fulfills a key assignment requirement. During installation, on the "Who are you?" screen, I created my user account with:

- **name:** Hiwot Adane
- **Computer name:** hiwot-adane-zorin
- **Username:** hiwotadane

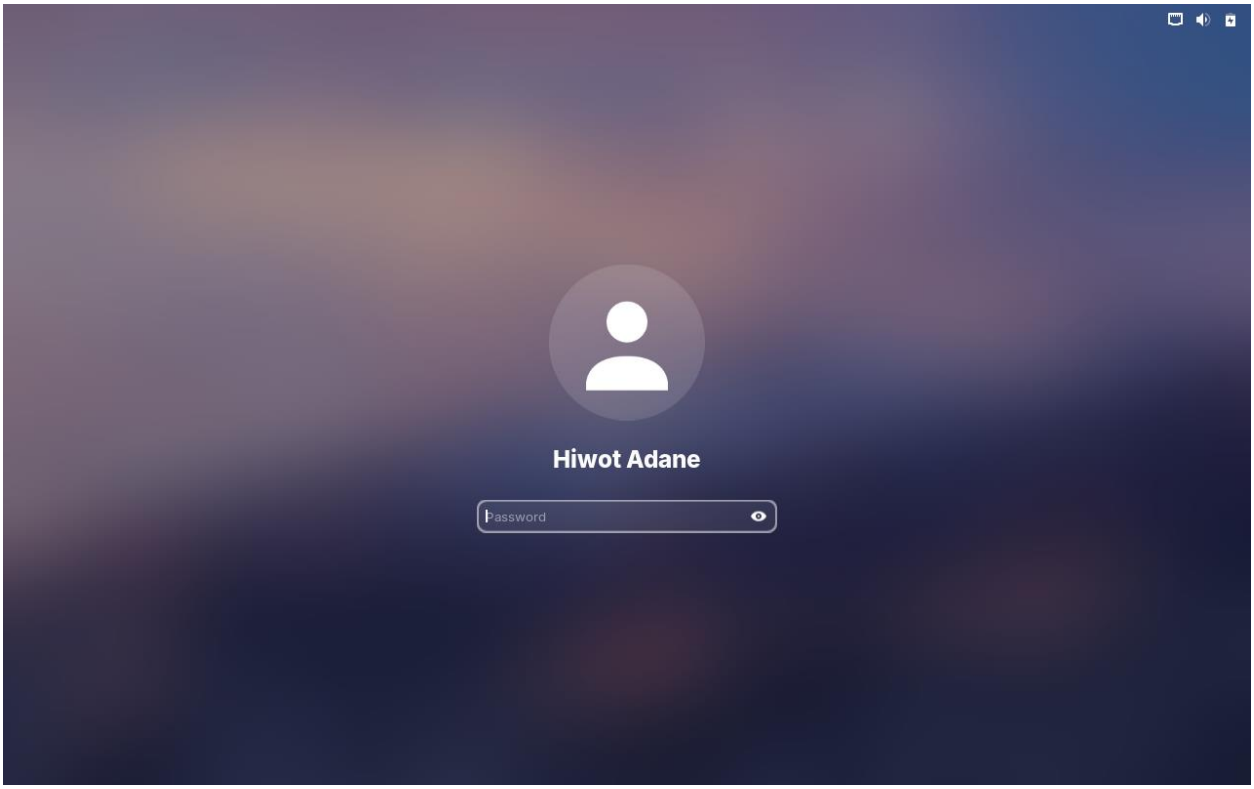


Figure12 : Zorin OS login screen showing my user account "Hiwot Adane" as required by the assignment

Step 13: Post-Installation — Welcome to Zorin OS

After logging in for the first time, the **"Welcome to Zorin OS 18.1"** screen appeared, offering a tour of features.

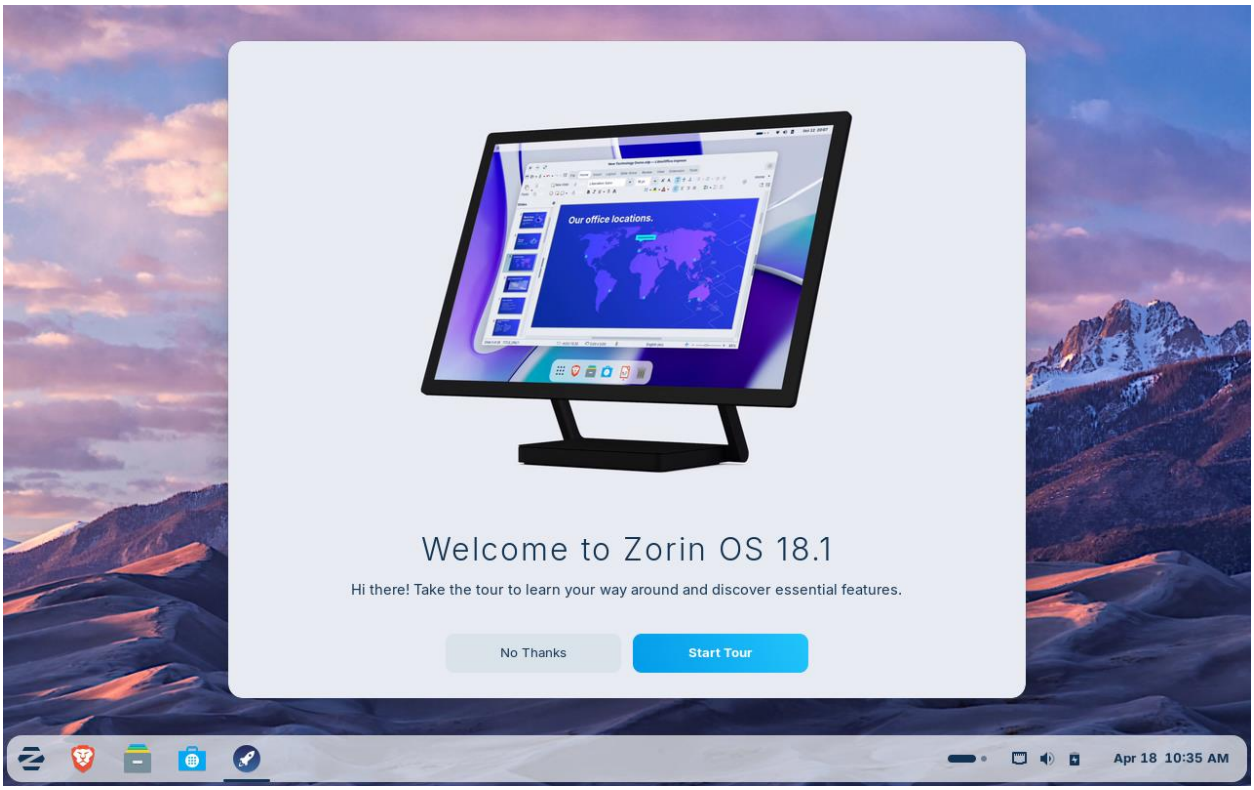


Figure 13: Welcome screen upon first login to Zorin OS 18.1

E, Issues

Issues	Solutions
Virtual machine not starting	Enable virtualization in BIOS
Slow performance	Reinstall VirtualBox or update it
Black screen after boot	Check ISO file integrity
ISO not detected	Install Guest Additions for better performance
"VT-x/AMD-V not enabled" error	Allocate more RAM/CPU cores
Virtual machine was very laggy	Increased RAM from 2 GB → 4 GB, allocated 2 CPU cores instead of 1
Screen resolution stuck at 800×600	Installed **VirtualBox Guest Additions**
Mouse cursor trapped inside VM window	Pressed **Right Ctrl** key to release mouse. After installing Guest Additions, mouse integration worked seamlessly
No sound in Zorin OS VM	In VM Settings → Audio → changed Audio Controller to **Intel HD Audio** → restarted VM
VM crashed once during installation	Re-created the VM from scratch and ensured the host PC had at least 30 GB free disk space
Network not working inside VM	In VM Settings → Network → changed from NAT to **Bridged Adapter** → restarted VM

Important Lesson: Installing **VirtualBox Guest Additions** solves multiple problems at once (display resolution, mouse integration, clipboard sharing, drag-and-drop, and time synchronization). This should be done immediately after installing any Linux virtual machine.

F, File System Support

What is a File System?

A file system is the method an operating system uses to store, organize, retrieve, and manage files on a storage device (hard disk, SSD, USB drive). Different operating systems use different file systems by default.

File Systems Supported by Zorin OS 18.1 (Linux-based)

File System	Read/Write Support in Zorin OS	Typical Use Case
ext4	Native (Default)	**Linux system drives –** chosen by Zorin OS during my installation

NTFS	Read/Write (via ntfs-3g driver)	Windows drives, external hard drives for cross-platform compatibility
FAT32	Read/Write	USB drives, SD cards – universal but individual file size limited to 4 GB
exFAT	Read/Write	Modern USB drives and SDXC cards – no 4 GB file limit
Btrfs	Supported (optional, not default)	Advanced Linux users – features: snapshots, compression, RAID
XFS	Supported	Large file servers, enterprise environments with very large files
ZFS	Available (needs manual installation)	High-end storage servers – data integrity, volume management
F2FS	Supported	Flash storage (SSDs, SD cards) – optimized for NAND flash memory
HFS+	Read-only (write support limited)	Older macOS drives
APFS	Not natively supported	Apple's newer file system – limited Linux support

Why Did Zorin OS Choose ext4 for My Installation?

During my installation, when I selected **"Erase disk and install Zorin OS,"** the installer automatically partitioned the virtual hard disk with **ext4** as the root file system. Here's why:

Reason	Explanation
Maturity	ext4 has been stable since 2008 – it's battle-tested and reliable
Performance	Excellent for general-purpose computing (fast boot times, quick file access)
Journaling	Protects against data corruption after system crashes or power loss
Backward compatibility	Can mount ext2 and ext3 drives without any

	issues
Large file support	Supports individual files up to 16 TB, volumes up to 1 exabyte (EB)
Open source	No licensing restrictions – unlike ZFS which has CDDL/GPL compatibility concerns
Default on most Linux distros	Ubuntu, Debian, Fedora, Linux Mint, and Zorin all default to ext4

G. Advantages and Disadvantages

Advantages of Installing Zorin OS in a Virtual Machine

Advantage	My Experience
No risk to host operating system	The host PC (my uncle's Windows computer) remained completely untouched. Even if I accidentally deleted system files inside Zorin, the host was safe.
Snapshots	Before trying something risky (like partitioning or installing unfamiliar software), I could take a snapshot. If something broke, I reverted in seconds.
Run both operating systems simultaneously	I could run Zorin OS in a window while keeping Windows open for research, taking notes, and capturing screenshots.
Portable	I can copy the VM folder (`C:\Users\Dell\VirtualBox VMs\Hiwot Adane`) to another computer (like my own) and run the exact same Zorin installation.
No dual-boot hassle	No need to restart the computer to switch between Windows and Zorin. No risk of bootloader corruption.
Practice without real hardware	Learned partitioning, file systems, Linux commands, and OS installation without buying a second computer.
Cost effective	VirtualBox is completely free. Zorin OS is free. No extra hardware costs.

Disadvantages of Installing Zorin OS in a Virtual Machine

Disadvantage	My Experience
Performance overhead	Zorin OS in a VM feels slightly slower than if installed directly on physical hardware. Boot time was ~30 seconds vs ~15 seconds bare-metal.
Limited graphics performance	3D games and heavy graphics software (like video editing or CAD) struggle because the virtual GPU is emulated. Guest Additions helps but doesn't fully solve this.
Shared resources	When the host PC (Windows) was busy (antivirus scan, Windows updates, multiple browser tabs), Zorin became laggy because both compete for the same CPU and RAM.
Host dependency	If Windows crashes or restarts, Zorin OS shuts down immediately – any unsaved work inside the VM is lost.
Storage overhead	The `.vdi` virtual disk file takes up real space on the physical SSD. At 30 GB allocated, about 12-15 GB was actually used after installation.
USB passthrough can be tricky	Plugging a USB drive directly into Zorin OS required extra configuration in VirtualBox settings (USB filters). Not as seamless as physical hardware.
Requires hardware virtualization support	The host CPU must support Intel VT-x or AMD-V. Older computers may not have this, making virtualization extremely slow or impossible.

H. Conclusion

The installation of Zorin OS in a virtual environment was successfully completed using VirtualBox. Despite encountering several technical issues, all problems were resolved, and the system is now fully operational. This assignment provided practical experience in virtualization and OS installation.

Key Takeaways for Me as a Student:

1. **Virtualization is an essential skill** for IT professionals. It allows safe experimentation with operating systems without risking physical hardware.
2. **Problems are normal** during OS installation – troubleshooting is a valuable part of the learning process. I encountered lag, resolution issues, mouse trapping, sound problems, and network configuration challenges – all solvable.

3. **Understanding file systems matters** because it affects performance, compatibility, and data safety. Zorin OS uses **ext4** for good reasons: stability, performance, journaling, and open-source licensing.
4. **Zorin OS is a beautiful, beginner-friendly Linux distribution** that runs well in a VM with proper resource allocation (4 GB RAM, 2 CPU cores, 128 MB video memory).
5. **VirtualBox Guest Additions is essential** – it solves display, mouse, clipboard, and time synchronization issues in one installation.

I. Future Outlook (Recommendations)

- **Practice installing other operating systems** in VMs (Windows 11, FreeBSD, Rocky Linux, Arch Linux)
- **Learn Vagrant** – a tool that automates VM creation from configuration files (infrastructure as code)
- **Explore WSL2 (Windows Subsystem for Linux)** as a lighter alternative to full VMs on Windows – runs a Linux kernel directly inside Windows without emulation
- **Learn Docker** – understand how containers differ from VMs and when to use each
- Increase RAM for better performance
- Use SSD storage for faster installation
- Explore other operating systems
- Learn advanced virtualization tools
- Practice cloud-based virtualization

PART 2:

What is Virtualization?

Virtualization is a technology that creates **virtual versions** of physical computing resources – including CPUs, memory, storage, and network interfaces – so that multiple operating systems can run **simultaneously and independently** on a single physical machine.

Each virtual machine (VM) believes it has its own dedicated hardware (CPU cores, RAM, hard disk, network card), but in reality, a software layer called the **hypervisor** (or Virtual Machine Monitor) manages the sharing of physical resources among all running VMs

Why Use Virtualization?

Reason	Explanation	Real-World Example
Server consolidation	Run 10, 20, or even 100+ VMs on one physical server instead of buying 100 separate	A cloud provider like AWS runs thousands of customer VMs on a single physical

	computers	server, saving electricity, space, cooling, and maintenance costs
Isolation	A crash, virus, or security breach in one VM does not affect other VMs or the host operating system	Security researchers test malware inside VMs – if the VM gets infected, they simply delete it. The host OS stays completely clean.
Legacy software support	Run old operating systems that no longer support modern hardware	A bank runs Windows XP inside a VM on Windows Server 2022 to run legacy terminal software that only works on XP
Development and testing	Test your application on multiple operating systems from a single computer	A web developer tests their app in different browsers across Windows 11, Ubuntu Linux, and macOS VMs – all on the same laptop
Disaster recovery	VMs can be backed up, copied, and restored much faster than physical servers	After a server failure, restore the VM from backup to new hardware in minutes, not hours or days
Cloud computing	Amazon EC2, Microsoft Azure, and Google Cloud all sell virtualization to customers	You rent a VM in the cloud for $0.10/hour instead of buying a $5,000 physical server that you might only need for a few hours
Training and education	Students can practice installing and configuring operating systems without risk	**This assignment!** I installed Zorin OS in a VM without touching or damaging my Windows installation
Snapshots and rollback	Save the complete state of a VM and revert instantly	Before installing a major update or trying a risky configuration, take a 10-second snapshot. If something breaks, revert in 10 seconds.

How Does Virtualization Work in Modern Operating Systems?

The Hypervisor (Virtual Machine Monitor)

The hypervisor is the core software that enables virtualization. There are two main types:

Type	Name	Description	Examples
Type 1	Bare-metal hypervisor	Runs directly on	VMware ESXi,

		physical hardware (no host operating system). Has direct access to all hardware resources. Used by Data centers, cloud providers, enterprise servers	Microsoft Hyper-V, KVM (Linux), Xen
Type 2	Hosted hypervisor	Runs as an application on top of an existing operating system (Windows, macOS, Linux). The host OS manages hardware access. Used by Students, developers, desktop users	**Oracle VM VirtualBox** (what I used), VMware Workstation Pro, QEMU, Parallels Desktop

In my assignment: I used a **Type 2 hypervisor** (VirtualBox 7.0) running on **Windows 11** (host OS). VirtualBox is just another application on Windows – it requests resources (RAM, CPU time, disk access) from Windows, then creates the virtual environment for Zorin OS inside that application window.

Hardware-Assisted Virtualization (Intel VT-x / AMD-V)

Modern CPUs include special instruction sets that make virtualization faster, more efficient, and more secure:

Technology	Intel	AMD
CPU Virtualization	Intel VT-x	AMD-V
I/O Virtualization	Intel VT-d	AMD IOMMU

Without hardware-assisted virtualization: The hypervisor must translate every single CPU instruction in software (binary translation), which is extremely slow – often 10-50x slower than native execution.

With hardware-assisted virtualization: The CPU executes guest instructions almost directly (with minimal hypervisor intervention), achieving near-native performance (90-98% of bare-metal speed).

My Intel Core i5 processor has **VT-x**, which I verified was enabled in the BIOS before starting the installation. Without VT-x, the Zorin OS VM would have been unusably slow

Zorin OS cannot tell it's running on virtual hardware unless it specifically checks (for example, running the command `systemd-detect-virt` returns `oracle` for VirtualBox). All applications inside Zorin OS work normally.

Paravirtualization

Standard virtualization (called **full virtualization**) emulates real hardware devices completely in software. This is slow for some devices (especially disk I/O and networking).

Paravirtualization is a technique where the guest operating system uses special drivers that "know" they are running in a virtual environment. These drivers replace slow emulated devices with faster, more direct communication channels to the hypervisor.

Summary Table: Virtualization Fundamentals

Question	Answer
What is a hypervisor?	Software that creates and manages virtual machines
What is the difference between Type 1 and Type 2?	Type 1 runs directly on hardware (data centers). Type 2 runs on a host OS (desktops).
Why do we need VT-x/AMD-V?	Hardware acceleration for virtualization – without it, VMs are extremely slow
What are Guest Additions?	Paravirtualized drivers that improve performance and features in VMs
What is ext4?	The default Linux file system – journaling, high performance, open source
How is a VM different from a container?	VM has separate OS (GBs, minutes). Container shares host OS (MBs, seconds).